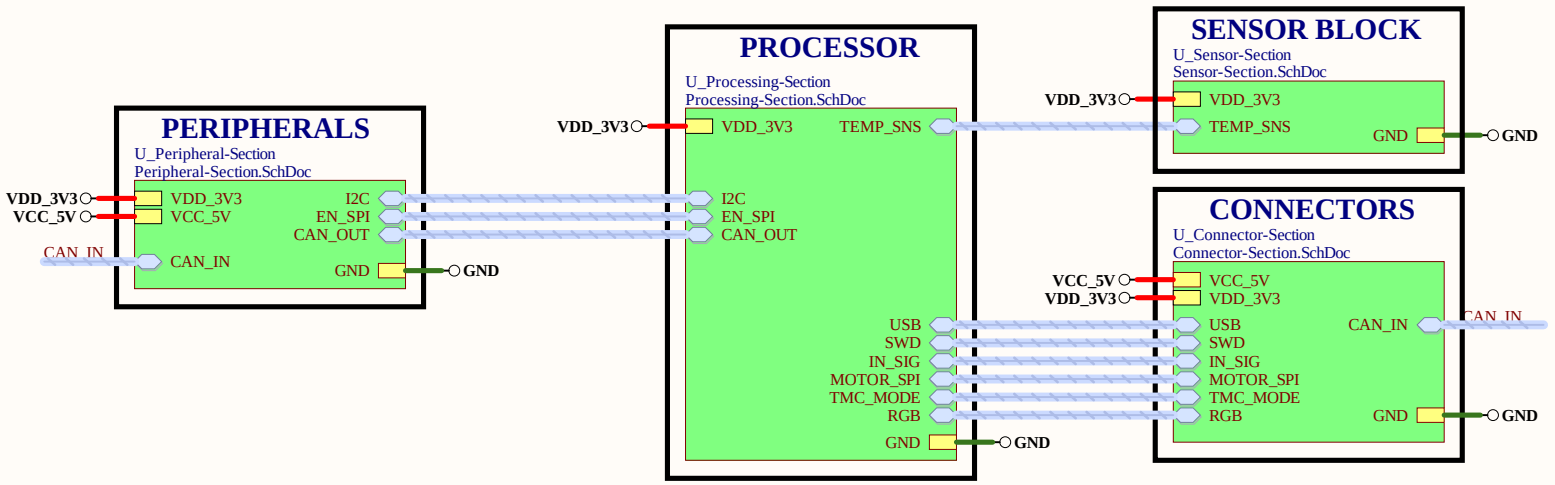
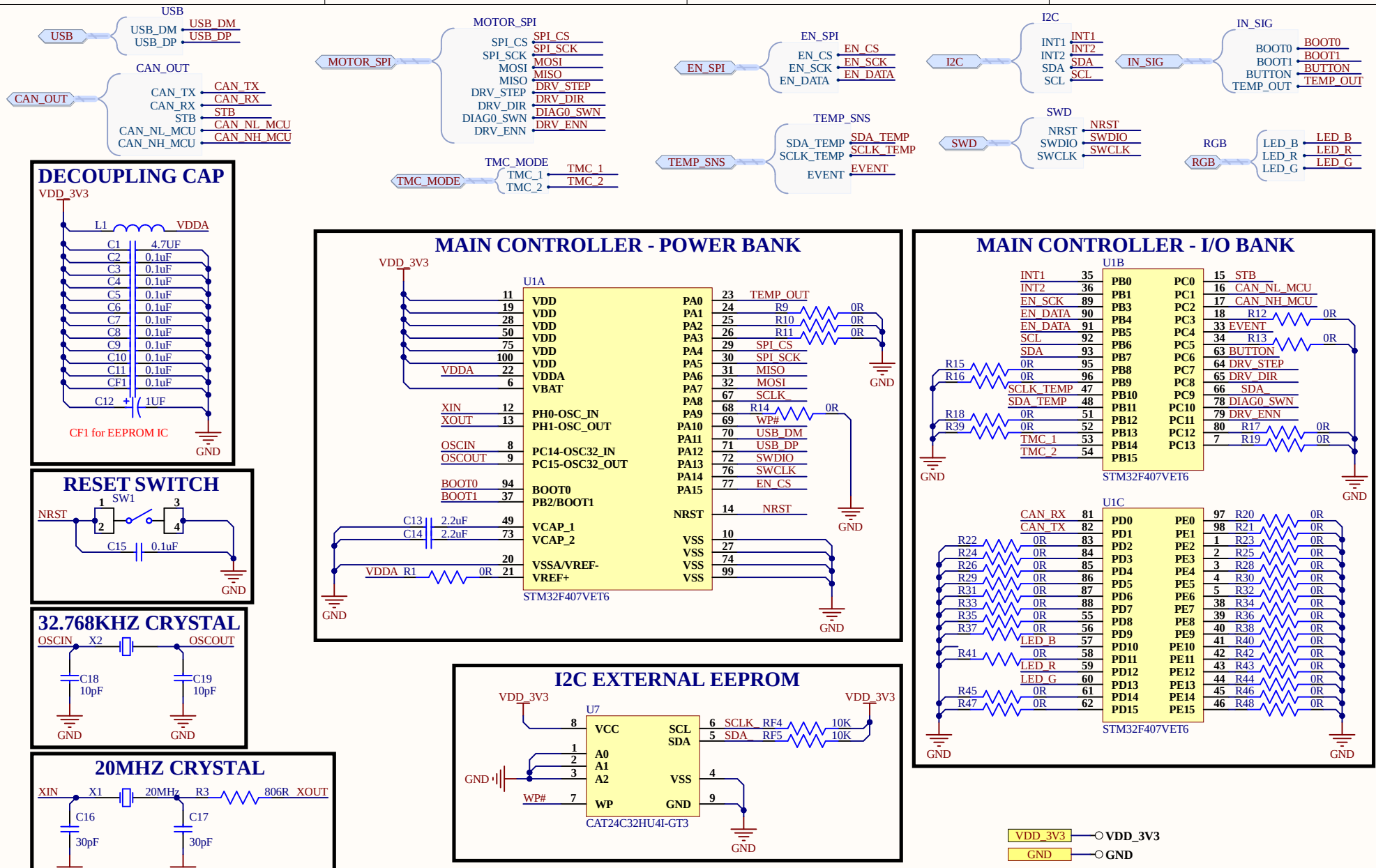
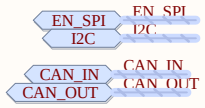
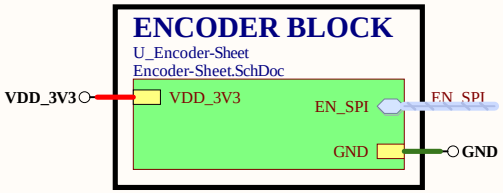
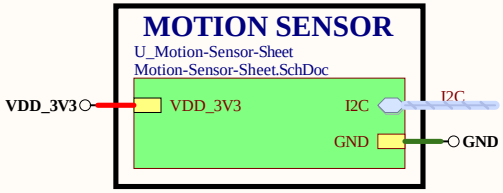
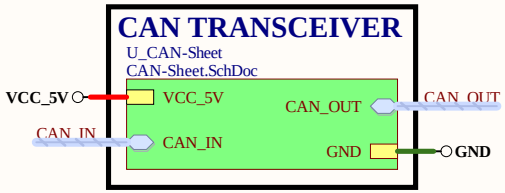

PERIPHERALS
U_Peripheral-Section
Peripheral-Section.SchDoc
VDD_3V3
VCC_5V
CAN_IN
VDD_3V3
VCC_5V
CAN_IN
I2C
EN_SPI
CAN_OUT
GND
GND
PROCESSOR
U_Processing-Section
Processing-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
I2C
EN_SPI
CAN_OUT
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
GND
GND
SENSOR BLOCK
U_Sensor-Section
Sensor-Section.SchDoc
VDD_3V3
VDD_3V3
TEMP_SNS
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VCC_5V
VDD_3V3
VCC_5V
VDD_3V3
USB
SWD
IN_SIG
MOTOR_SPI
TMC_MODE
RGB
CAN_IN
GND
GND
CAN_IN

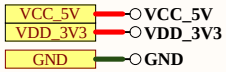

USB
USB
USB_DM
USB_DP
CAN_OUT
CAN_OUT
CAN_TX
CAN_RX
STB
CAN_NL_MCU
CAN_NH_MCU
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TMC_MODE
TMC_1
TMC_2
EN_SPI
EN_CS
EN_SCK
EN_DATA
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
I2C
INT1
INT2
SDA
SCL
SWD
NRST
SWDIO
SWCLK
IN_SIG
BOOT0
BOOT1
BUTTON
TEMP_OUT
RGB
LED_B
LED_R
LED_G
DECOUPLING CAP
VDD_3V3
L1
VDDA
C1 4.7UF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
C7 0.1uF
C8 0.1uF
C9 0.1uF
C10 0.1uF
C11 0.1uF
CF1 0.1uF
C12 1UF
CF1 for EEPROM IC
GND
RESET SWITCH
NRST
SW1
C15 0.1uF
GND
32.768KHZ CRYSTAL
OSCIN
X2
OSCOUT
C18 10pF
C19 10pF
GND
20MHZ CRYSTAL
XIN
X1
20MHz
R3 806R
XOUT
C16 30pF
C17 30pF
GND
MAIN CONTROLLER - POWER BANK
VDD_3V3
U1A
VDD
VDDA
VBAT
PH0-OSC_IN
PH1-OSC_OUT
PC14-OSC32_IN
PC15-OSC32_OUT
BOOT0
PB2/BOOT1
VCAP_1
VCAP_2
VSSA/VREF-
VREF+
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
NRST
VSS
TEMP_OUT
R9 0R
R10 0R
R11 0R
SPI_CS
SPI_SCK
MISO
MOSI
SCLK_
R14 0R
WP#
USB_DM
USB_DP
SWDIO
SWCLK
EN_CS
NRST
C13 2.2uF
C14 2.2uF
VDDA_R1 0R
STM32F407VET6
GND
MAIN CONTROLLER - I/O BANK
U1B
U1C
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
STB
CAN_NL_MCU
CAN_NH_MCU
EVENT
BUTTON
DRV_STEP
DRV_DIR
SDA_
DIAG0_SWN
DRV_ENN
STM32F407VET6
GND
I2C EXTERNAL EEPROM
VDD_3V3
U7
VCC
A0
A1
A2
WP
SCL
SDA
VSS
GND
SCLK_ RF4 10K
SDA_ RF5 10K
WP#
CAT24C32HU4I-GT3
GND
VDD_3V3
VDD_3V3
GND
GND


EN_SPI
I2C
EN_SPI
I2C
CAN_IN
CAN_OUT
CAN_IN
CAN_OUT

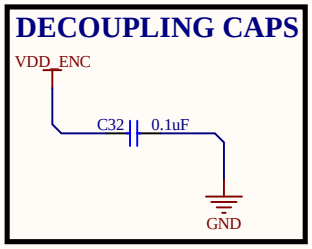

ENCODER BLOCK
U_Encoder-Sheet
Encoder-Sheet.SchDoc
VDD_3V3
VDD_3V3
EN_SPI
EN_SPI
GND
GND

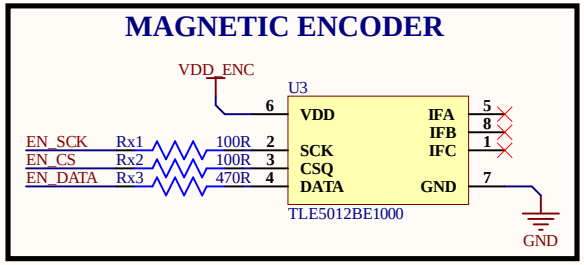

MOTION SENSOR
U_Motion-Sensor-Sheet
Motion-Sensor-Sheet.SchDoc
VDD_3V3
VDD_3V3
I2C
I2C
GND
GND

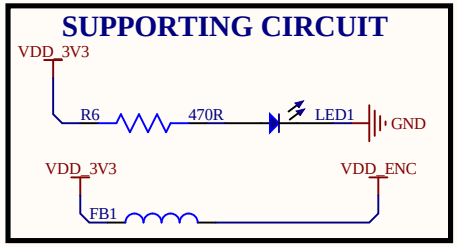

CAN TRANSCEIVER
U_CAN-Sheet
CAN-Sheet.SchDoc
VCC_5V
VCC_5V
CAN_OUT
CAN_OUT
CAN_IN
CAN_IN
GND
GND

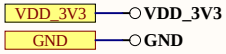

VCC_5V
VDD_3V3
GND
VCC_5V
VDD_3V3
GND

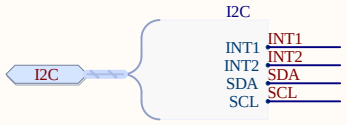

EN_SPI
EN_SPI
EN_CS
EN_SCK
EN_DATA
EN_CS
EN_SCK
EN_DATA

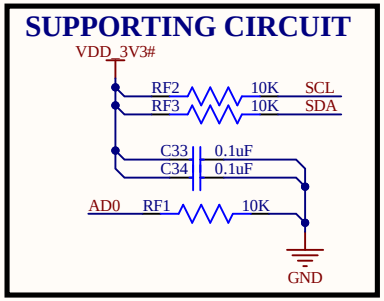

DECOUPLING CAPS
VDD_ENC
C32
0.1uF
GND

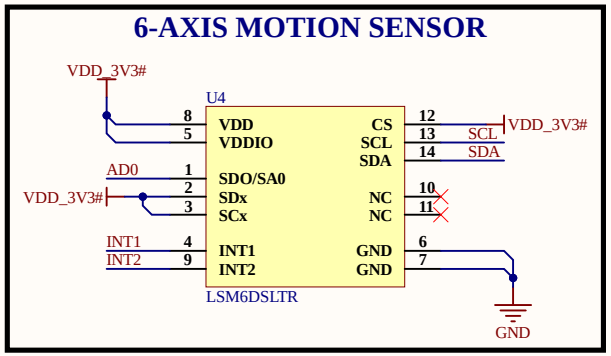

MAGNETIC ENCODER
VDD_ENC
U3
6
VDD
IFA
5
IFB
8
IFC
1
EN_SCK
Rx1
100R
2
SCK
EN_CS
Rx2
100R
3
CSQ
EN_DATA
Rx3
470R
4
DATA
GND
7
TLE5012BE1000
GND

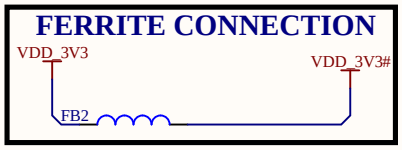

SUPPORTING CIRCUIT
VDD_3V3
R6
470R
LED1
GND
VDD_3V3
VDD_ENC
FB1

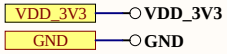

VDD_3V3
VDD_3V3
GND
GND

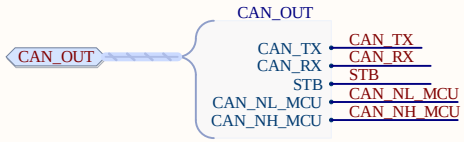

I2C
I2C
INT1
INT2
SDA
SCL
INT1
INT2
SDA
SCL

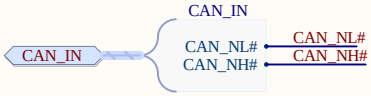

SUPPORTING CIRCUIT
VDD_3V3#
RF2
10K
SCL
RF3
10K
SDA
C33
0.1uF
C34
0.1uF
AD0
RF1
10K
GND


6-AXIS MOTION SENSOR
VDD_3V3#
U4
8
5
VDD
VDDIO
CS
SCL
SDA
12
13
14
VDD_3V3#
SCL
SDA
AD0
1
SDO/SA0
VDD_3V3#
2
3
SDx
SCx
NC
NC
10
11
INT1
INT2
4
9
INT1
INT2
GND
GND
6
7
LSM6DSLTR
GND

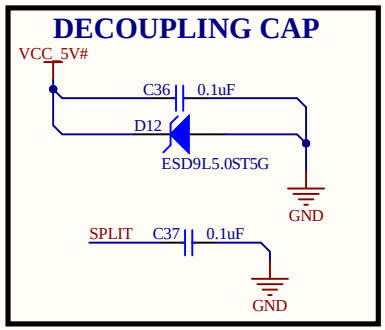

FERRITE CONNECTION
VDD_3V3
VDD_3V3#
FB2

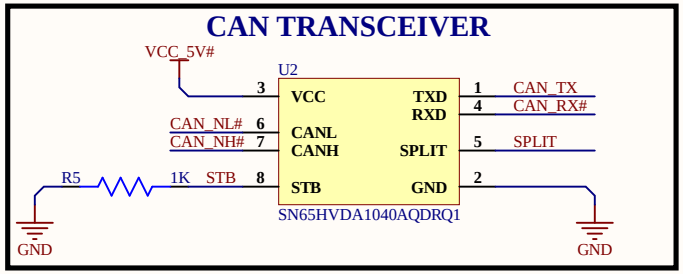

VDD_3V3
VDD_3V3
GND
GND

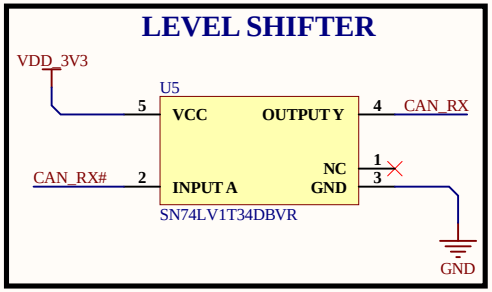


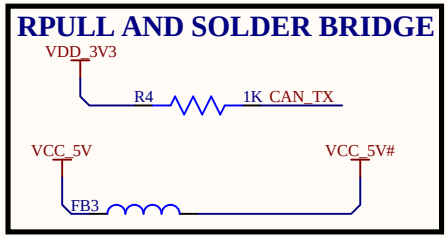


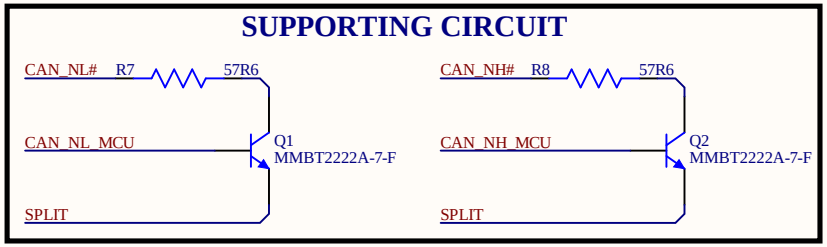
Connections for CAN are w.r.t Controller Rx and Tx

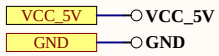


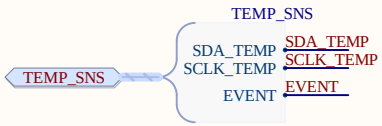


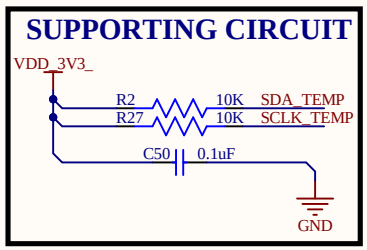


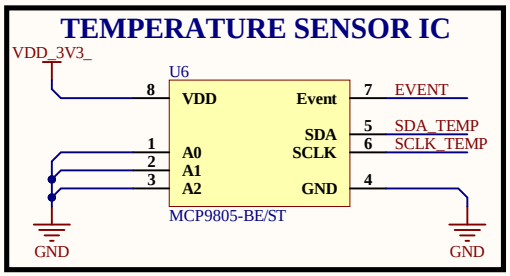


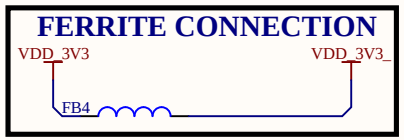


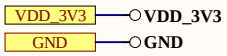


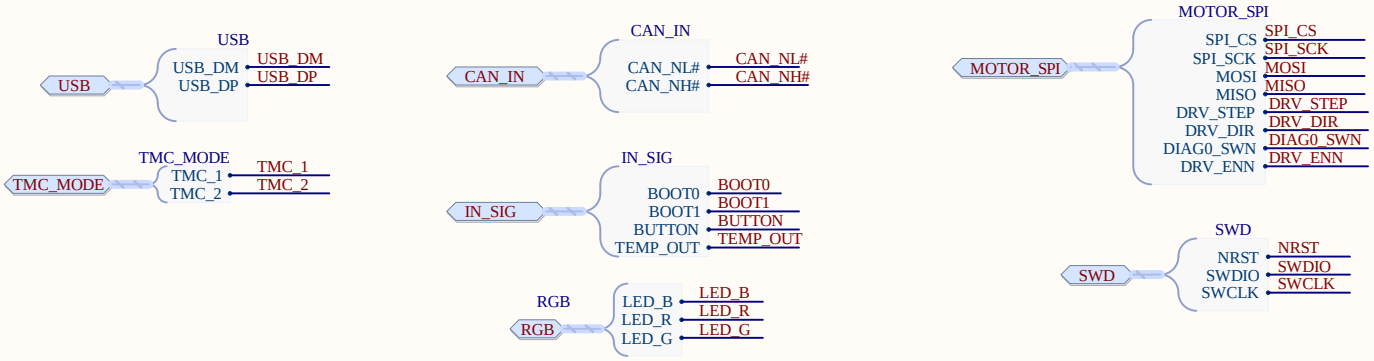

TEMP_SNS
TEMP_SNS
SDA_TEMP
SCLK_TEMP
EVENT
SDA_TEMP
SCLK_TEMP
EVENT


SUPPORTING CIRCUIT
VDD_3V3_
R2
10K
SDA_TEMP
R27
10K
SCLK_TEMP
C50
0.1uF
GND

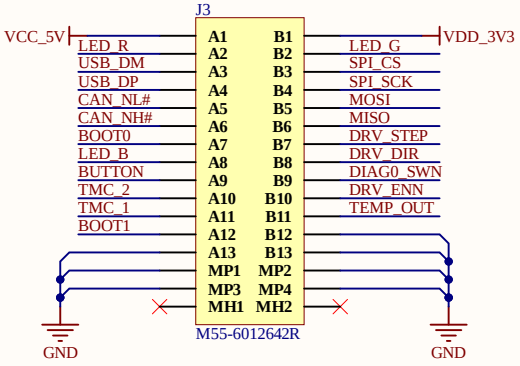

TEMPERATURE SENSOR IC
VDD_3V3_
U6
8
VDD
Event
7
EVENT
SDA
5
SDA_TEMP
1
A0
SCLK
6
SCLK_TEMP
2
A1
3
A2
GND
4
MCP9805-BE/ST
GND
GND


FERRITE CONNECTION
VDD_3V3
VDD_3V3_
FB4

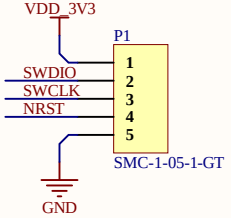

VDD_3V3
VDD_3V3
GND
GND



BOARD TO BOARD CONNECTOR



PROGRAMMING HEADER



TVS PROTECTION DIODES

